

1. 需求分析

1.1 系统需求

1.1.1 用户注册

```
flowchart TD
    Start([开始]) --> InputCode{输入验证码}
    InputCode --> VerifyCode{验证验证码}
    InputCode --> Register[注册]
    VerifyCode --> BindUser[绑定用户]
    VerifyCode --> ShowError[显示错误]
    BindUser --> UpdateCode[更新验证码]
```

1.2 系统架构

1.2.1 系统架构

```
stateDiagram-v2
    [*] --> Admin: Admin 登录
    Admin --> Admin: 退出
    Admin --> Admin: 修改密码
    Admin --> Admin: 修改昵称
    Admin --> Admin: use_count >= max Uses
    Admin --> Admin: expires_at
    Admin --> Admin: Admin 退出
    Admin --> [*]
    Admin --> [*]
```

1.3 数据库设计

1.3.1 数据库设计

```
erDiagram
    users ||--o{ referral_codes : "referral_codes : (referrer_id)"
    users ||--o{ referral_usage_logs : "referral_usage_logs"
    users ||--o{ referral_relationships : "referral_relationships : (referrer_id)"
    users ||--o{ referral_relationships : "referral_relationships : (referee_id)"
    users ||--o{ commission_records : "commission_records : (user_id)"
    referral_codes ||--o{ referral_usage_logs : "referral_usage_logs : (code)"
    referral_codes {
        int id PK
    }
```

1.4 接口设计

1.4.1 接口设计

```
sequenceDiagram
    participant Admin as Admin
    participant Frontend as Frontend
    participant Backend as Backend API
    participant DB as DB
    Admin->>Frontend: 获取验证码
    Frontend->>Backend: GET /api/admin/referral-codes/list
    Backend->>DB: SELECT * FROM referral_codes
    DB-->>Backend: 返回验证码
```

1.5 系统部署

[5 -]

```
flowchart TD
    Start([ ]) --> TriggerType{ }
    TriggerType -->| | Calc1[ ]
    TriggerType -->| | Calc2[ ]
    TriggerType -->| | Calc3[ ]
    TriggerType -->| | Calc4[ ]
    Calc1 --> GetReferral[ ]
    Calc2 --> GetReferral
```

1.

- **`GYRG` + 4
- **`LCRG` + 4
- **`ZYRG` + 4
- **`SQZD` + 4
- **`CSHH` + 4

2.

- **Level 1**A B
- **Level 2**A B B C A C

3.

- **T+1**/pending
- **T+7**confirmed
- **15** confirmed paid

4.

- IP 24 3
- 7
-

SQL

```
WITH RECURSIVE team AS (
-- 
SELECT
rr.referee_id,
rr.referrer_code,
rr.referee_code,
rr.level,
```

```

1 as depth
FROM referral_relationships rr
WHERE rr.referrer_id = ? -- ██████ID

UNION ALL

-- ██████████
SELECT
rr.referee_id,
rr.referrer_code,
rr.referee_code,
rr.level,
t.depth + 1
FROM referral_relationships rr
INNER JOIN team t ON rr.referrer_id = t.referee_id
WHERE t.depth < 5 -- ███5█
)
SELECT * FROM team;

```

██████████

```

SELECT
u.id,
u.nickname,
rc.code as referral_code,
COUNT(DISTINCT rr.referee_id) as total_referees,
COUNT(DISTINCT CASE WHEN u_reg.created_at >= date('now', '-30 days') THEN rr.referee_id END) as
month_new,
COALESCE(SUM(cr.amount), 0) as total_commission
FROM users u
LEFT JOIN referral_codes rc ON rc.referrer_id = u.id
LEFT JOIN referral_relationships rr ON rr.referrer_id = u.id
LEFT JOIN commission_records cr ON cr.user_id = u.id AND cr.status = 'paid'
WHERE u.role IN ('creator', 'professional', 'public_welfare')
GROUP BY u.id;

```